ChatGPT

L21_L22-36-129.pdf
PDF

Create a complete lecture scribe using only the provided material.

Follow the lecture strictly from first to last slide. Do not skip anything.

Include every element exactly as given: all definitions, statements, theorems, proofs, derivations, equations, examples, figures, cases of real life application and any other content. Write all steps fully.

Do not modify, summarize, interpret, or add anything.

Keep the following structure EXACTLY the same and do not change it:

Example Lecture Scribe (structure only, do not copy content):

Topic Title

Overview
Brief statement describing what the lecture covers.

Definitions and Notation
Formal definitions and symbols introduced in the lecture.

Main Results
Statements of theorems, propositions, or key results, section-wise.
Include figures, graphs, tables, or diagrams ONLY if they explicitly appear in the lecture.
If a figure is referenced in the lecture, include a formal textual description exactly as stated.

Derivations / Proofs
Step-by-step derivations or proofs exactly as presented.

Worked Examples
Example problems solved in class with all intermediate steps.

Summary
Concise recap strictly limited to formulas, results, and definitions already stated.

Below is the lecture scribe constructed strictly from the provided slides L21_L22-36-129, following the required structure and preserving content exactly as presented.

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## Topic Title

Chebyshev's Inequality, Chernoff Bounds, and Jensen's Inequality

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## Overview

The lecture covers probabilistic tail bounds including Markov, Chebyshev, and Chernoff bounds, followed by Jensen's Inequality and its connection to convexity and expectations.

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## Definitions and Notation

- **Markov Inequality:**
  $P(X \geq a) \leq \frac{E[X]}{a}$
- **Chebyshev Inequality:**
  $P(|X - \mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$
- **Random Variable (Binomial):**
  $X \sim \mathrm{Binomial}(n, \frac{1}{2})$,
  $\mu = \frac{n}{2}, \quad \mathrm{Var}(X) = \frac{n}{4}$
- **Convex Function Definition:**
  A function $g(x)$ is convex on interval $S$ if, for any $x_1, x_2 \in S$ and $\alpha \in [0, 1]$,
  $g(\alpha x_1 + (1-\alpha)x_2) \leq \alpha g(x_1) + (1-\alpha)g(x_2)$
- **Expectation:**
  $E[X] = \sum p_X(x_i)x_i$
- **Expectation of function:**
  $E[g(X)] = \sum p_X(x_i)g(x_i)$

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## Main Results

### Chebyshev's Inequality

- Uses mean + variance
- Controls deviation from mean
- $P(|X - \mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$
- **Proof Idea:**
  Apply Markov on $(X - \mu)^2$:

$$P(|X - \mu| \geq a) = P((X - \mu)^2 \geq a^2) \leq \frac{E[(X - \mu)^2]}{a^2} = \frac{\mathrm{Var}(X)}{a^2}$$

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### Running Example (Binomial)

$$P\left(X \geq \frac{3}{4}n\right) = P\left(|X - \frac{n}{2}| \geq \frac{n}{4}\right) \leq \frac{n}{(n/4)^2} = \frac{4}{n}$$

Key Insight: Bound decreases as $n$ increases

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### Real-Life Example

- Mean = 70
- Std deviation = 10

Question: fraction below 40 or above 100

$$P(|X - 70| \geq 30) \leq \frac{10^2}{30^2} = \frac{1}{9}$$

At most $\frac{1}{9}$ students are far from mean

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### Solved Example 1

- $\mu = 70, \sigma = 5 \Rightarrow \sigma^2 = 25$
- $X \geq 90 \Rightarrow |X - 70| \geq 20$

$$P(X \geq 90) \leq P(|X - 70| \geq 20) \leq \frac{25}{400} = \frac{1}{16} = 0.0625$$

Result: Much tighter than Markov

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### Solved Example 2 (Network Latency)

- $\mu = 50, \sigma^2 = 16$

$$P(|X - 50| \geq 20) \leq \frac{16}{400} = \frac{1}{25}$$

Result: At most 4% packets outside range

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### Chernoff Bound: Key Idea

Progression:

- Markov → $X$
- Chebyshev → $(X - \mu)^2$
- Chernoff → $e^{tX}$

Key Insight:

- Exponential grows very fast
- Tail events get amplified
- Better transformation ⇒ tighter tail bound

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### Core Trick

$$P(X \geq a) = P(e^{tX} \geq e^{ta})$$

Apply Markov:

$$P(e^{tX} \geq e^{ta}) \leq \frac{E[e^{tX}]}{e^{ta}}$$

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### Final Chernoff Bound

$$P(X \geq a) \leq \min_{t>0} \left\{ \frac{E[e^{tX}]}{e^{ta}} \right\}$$

Key Insight:

- Bound holds for all $t > 0$
- Choose best $t$ ⇒ tightest bound
- Exponential decay instead of polynomial

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### Lower Tail Chernoff

$$P(X \leq a) \leq \min_{t<0} \left\{ \frac{E[e^{tX}]}{e^{ta}} \right\}$$

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### Poisson Chernoff Bound

$$E[e^{tX}] = \sum_{i=0}^{\infty} e^{ti} \frac{e^{-\lambda} \lambda^i}{i!} = e^{\lambda(e^t - 1)}$$

$$P(X \geq a) \leq \min_{t>0} e^{\lambda(e^t-1) - ta}$$

Minimized at $t = \ln(a/\lambda)$

Thus:

$$P(X \geq a) \leq e^{a-\lambda} \left( \frac{\lambda}{a} \right)^a$$

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### Chernoff Example (Binomial Lower Tail)

- $X \sim \text{Binomial}(20, 0.5), \mu = 10$
- $\delta = 1$

$$P(X \leq 0) \leq \exp\left( -\frac{\mu\delta^2}{2} \right) = e^{-5} \approx 0.0067$$

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### Jensen's Inequality

We know:

$$E[X^2] \geq (E[X])^2$$

General Result:

$$E[X^a] \geq (E[X])^a$$

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### Convexity

- Curve stays below line between points
- $g''(x) \geq 0$

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### From Convexity to Expectation

Convex combination:

$$g\left( \sum \alpha_i x_i \right) \leq \sum \alpha_i g(x_i)$$

Interpretation:

- $\alpha_i$ act like probabilities
- Weighted average

Thus:

$$g(E[X]) \leq E[g(X)]$$

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### Jensen's Theorem

If $g(x)$ is convex and $X \in S$:

$$g(E[X]) \leq E[g(X)]$$

Interpretation:

- Average first ⇒ smaller value
- Apply function first ⇒ larger value
- Convex functions push expectation upwards

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### Jensen Simple Example

$X = \{1, 3\}$ with probability 0.5

$$E[X] = 2$$

$$g(E[X]) = 2^2 = 4$$

$$E[g(X)] = 0.5(1^2) + 0.5(3^2) = 5$$

Thus:

$$g(E[X]) \leq E[g(X)]$$

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### Jensen Solved Example

$X = 2, 4$

$$E[X] = 3$$

$$g(E[X]) = 9$$

$$E[g(X)] = 0.5(4) + 0.5(16) = 10$$

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## Derivations / Proofs

### Chebyshev Proof

$$P(|X-\mu| \geq a) = P((X-\mu)^2 \geq a^2) \leq \frac{E[(X-\mu)^2]}{a^2} = \frac{\mathrm{Var}(X)}{a^2}$$

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### Chernoff Proof Structure

1. Transform: $P(X \geq a) = P(e^{tX} \geq e^{ta})$
2. Apply Markov
3. Minimize over $t$

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### Jensen Proof Idea

From convexity:

$$g\left(\sum \alpha_i x_i\right) \leq \sum \alpha_i g(x_i)$$

Replace weights with probabilities ⇒ expectation form

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## Worked Examples

1. Exam Scores → $\frac{1}{9}$ bound
2. Grades variance → $\frac{1}{16}$
3. Network latency → $\frac{1}{25}$
4. Binomial Chernoff → $e^{-5} \approx 0.0067$
5. Jensen numeric verification → 4 ≤ 5

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## Summary

- Markov:
  $P(X \geq a) \leq \frac{E[X]}{a}$
- Chebyshev:
  $P(|X-\mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$
- Chernoff:

$$P(X \geq a) \leq \min_{t>0} \frac{E[e^{tX}]}{e^{ta}}$$

- Jensen:

$$g(E[X]) \leq E[g(X)]$$

- Key Ideas:
  - Tail bounds improve from Markov → Chebyshev → Chernoff
  - Exponential gives strongest bounds
  - Convexity connects to expectation via Jensen

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If you want, I can convert this into **perfect exam notes / handwritten style / LaTeX PDF**.

Sources

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